

COURSE 3 - DATA ENGINEERING 2026

C3-072 - Streaming & Event-Driven Pipelines

Protected Review

Open only after the matching Pxx attempt is saved. Repair guidance, not a solution bank.

Artifact	C3-072
Status	2026 - LEARNER EDITION
Phase	C3-06 Streaming & Event-Driven Pipelines 13 lessons 34-46 focused hours
Prerequisite	C3-05 Gate PASS; C2B streaming/lakehouse concepts may be recalled but Kafka/Spark streaming implementation is taught here.
Current lane	Apache Kafka 4.3.1 Spark Structured Streaming 4.2.0 Delta Lake 4.4.0 Debezium 3.6.2.Final
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny event example -> guided Kafka/streaming action -> expected result -> why it worked -> break/fix -> independent changed case -> validation -> replay/state/production reasoning -> evidence. A running stream is not mastery if duplicates, late data, offsets, checkpoints or sink state are wrong.

Protected-use rule

For one attempted task only: compare your evidence, identify the failed competency, close this file, then solve a changed task. Keep the original attempt and retry append-only.

R01 - Review for P01

What strong evidence shows

Architecture choice follows latency/replay/state/operating requirements; streaming is not automatically chosen.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
- Learner can explain the failure/recovery without reading this page

Targeted repair

If every workload becomes streaming, restate the business latency and replay requirement. A daily dashboard rarely needs second-level event processing.

Changed retry

Change one workload from "alert within 10 seconds" to "report by 8 AM" and re-evaluate the architecture.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R02 - Review for P02

What strong evidence shows

Ordering/key/partition/replication boundaries are correct and RF=1 is never presented as HA.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
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Targeted repair

If you assume one global offset or ordering across partitions, redraw offsets per partition.

Changed retry

Switch the key from order_id to random UUID and explain the consequence for per-order ordering.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R03 - Review for P03

What strong evidence shows

One group shares partitions; independent groups keep independent progress; scaling is bounded by partitions and rebalances.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
- Learner can explain the failure/recovery without reading this page

Targeted repair

If you say Kafka "sends each event to only one application," distinguish one consumer group from multiple independent groups.

Changed retry

Change the topic from 3 to 6 partitions and re-evaluate group parallelism and ordering.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R04 - Review for P04

What strong evidence shows

Committed offsets, retention and replay are separate concepts; replay cannot corrupt downstream state.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
- Learner can explain the failure/recovery without reading this page

Targeted repair

If replay duplicates output, repair the sink identity/dedupe rule rather than disabling replay.

Changed retry

Reduce simulated retention below the lag window and explain the data-loss risk.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R05 - Review for P05

What strong evidence shows

Schema changes are checked under an explicit compatibility policy and rollout plan; valid JSON alone is not enough.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
- Learner can explain the failure/recovery without reading this page

Targeted repair

If you equate "valid JSON" with "compatible event," test a consumer expecting v1 against v2.

Changed retry

Design v3 that adds an optional `campaign_id` while retaining replay compatibility.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R06 - Review for P06

What strong evidence shows

CDC flow includes operation/key/source position/delete/schema/restart responsibilities; connector plumbing is not confused with business correctness.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
- Learner can explain the failure/recovery without reading this page

Targeted repair

If you flatten CDC to "latest row only," identify what delete/history/replay information was lost.

Changed retry

Change one source update into delete + recreate and explain downstream semantics.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R07 - Review for P07

What strong evidence shows

Duplicate delivery converges to one business state with stable event identity; guarantee claims name their exact scope.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
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Targeted repair

If you dedupe on order_id, restore legitimate multiple lifecycle events.

Changed retry

Change the domain to payment attempts where repeated order_id can be legitimate and choose a new idempotency key.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R08 - Review for P08

What strong evidence shows

Event/ingest/processing clocks remain distinct and late arrival policy is explicit.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
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Targeted repair

If you classify late as invalid automatically, separate "late but valid" from malformed/impossible timestamp.

Changed retry

Change allowed lateness from 10 to 3 minutes and identify which business metric changes.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R09 - Review for P09

What strong evidence shows

Watermark/window reasoning predicts state and late-event behavior; timezone/event-time field are declared.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
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Targeted repair

If a late event surprises you, record the prior max event time, watermark and target window instead of guessing.

Changed retry

Repeat with 15-minute windows and a 5-minute watermark.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R10 - Review for P10

What strong evidence shows

Structured Streaming uses DataFrame APIs with explicit source/schema/sink/checkpoint and live query progress only when actually executed.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
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Targeted repair

If you use inferSchema on an unbounded JSON stream, move schema to code/contract. If output differs after rerun, inspect checkpoint identity.

Changed retry

Point the same transformation at a second inbox with a fresh checkpoint and compare results.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R11 - Review for P11

What strong evidence shows

Same checkpoint resumes the same query identity; fresh checkpoint means new/replay behavior; no checkpoint surgery.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
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Targeted repair

If a restart seems wrong, do not delete random checkpoint files. Restore the full checkpoint or intentionally start a new query and treat it as replay.

Changed retry

Change stateful window keys and explain why a new checkpoint/migration plan is safer.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R12 - Review for P12

What strong evidence shows

Bronze/Silver sinks use separate checkpoints and deterministic idempotent/transactional boundaries.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
- No fabricated Kafka/Spark/Delta runtime screenshot or metric
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Targeted repair

If Bronze and Silver share one checkpoint or overwrite raw event history, separate query identity and evidence-preserving storage.

Changed retry

Add a replay/backfill path for a historical offset range without corrupting live progress.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.

R13 - Review for P13

What strong evidence shows

Lag/rate/batch/state/watermark/sink metrics are tied to SLO/retention risk and a concrete recovery runbook.

Check your attempt

- Correct event identity and event-time meaning
- No hidden duplicate/loss/replay corruption
- Offset/checkpoint/state claim supported by evidence when runtime exists
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Targeted repair

If you alert on "lag > 0", tie thresholds to normal traffic, recovery time and retention headroom.

Changed retry

Change source traffic 5x while sink throughput stays fixed; calculate qualitative backlog direction and escalation action.

Close this review now. Save the changed retry separately; do not overwrite the failed attempt.